Physical Chemistry (II)

Lecture 03

Principles of Quantum Mechanics

In the Last Class

- Classical mechanics
  - Physics of particles
  - Physics of waves

- Old quantum theory
  - Wave-particle duality
  - Bohr atom: quantization
  - Introduction to spectroscopy

Physics of Particles

- Described by $\vec{r}$ and $\vec{v}$ (or equivalently $\vec{p} = m\vec{v}$)
- Kinetic energy $T = mv^2/2$ + potential energy V
- Dynamics: $\vec{F} = m\vec{a}$ or

$$\frac{\partial^2 \vec{r}}{\partial t^2} = -\frac{\nabla V}{m}$$

Physics of Waves

- Described by amplitude $A = A(\vec{r}, t)$

- Dynamics: wave equation

$$\frac{\partial^2 A}{\partial t^2} = u^2 \nabla^2 A$$

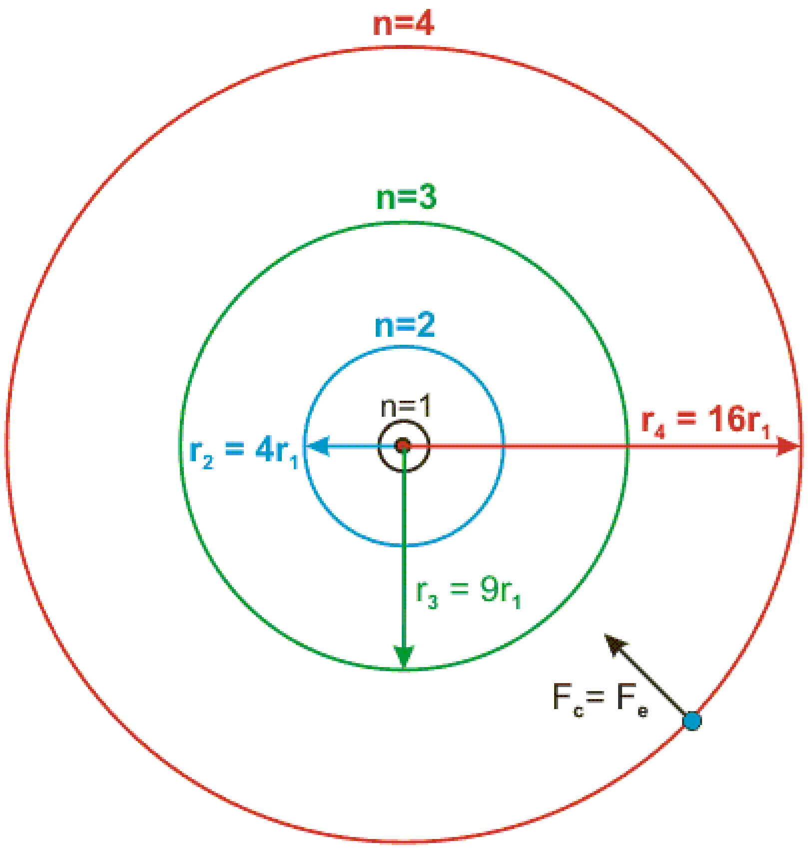
- Superposition of waves: $A(\vec{r}, t) = A_1(\vec{r}, t) + A_2(\vec{r}, t) + \cdots$

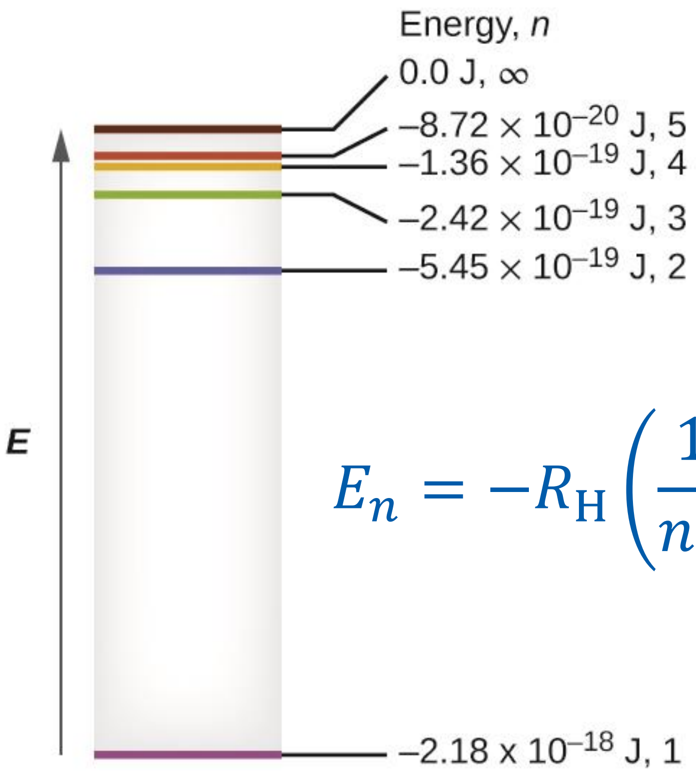
- Refraction, interference, diffraction

Wave-Particle Duality

- Particle nature of light
  - Photoelectric effect: $E = h\nu = hc/\lambda$
  - Compton scattering: $p = h/\lambda$
- Wave nature of matter
  - Matter wave: $\lambda = h/p = h/(mv)$
- Wave-particle duality
  - Light and matter exhibit both properties of waves and particles
  - Bohr's complementarity principle

Bohr Atomic Model



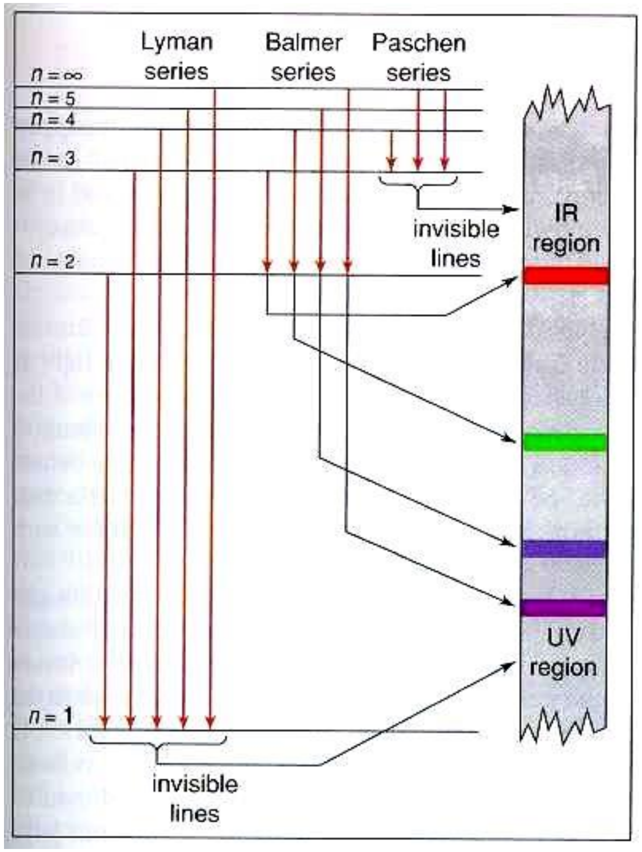


$$E_n = -R_{\mathrm{H}}\left(\frac{1}{n^2}\right)$$

(Image: https://www.webassign.net/question_assets/wertzcams3/ch_2/manual.html
https://opentextbc.ca/chemistry/chapter/6-2-the-bohr-model/)

Last Class

Transitions and Spectroscopy



(Image: https://bramblechemistry.weebly.com/4e2-bohr-model.html)

Where Are We?

기본 원리	응용
단일 입자 계	PIB, SHO, RR 수소 원자
다입자 계	나머지 모든 원자 모든 분자

Postulates of QM

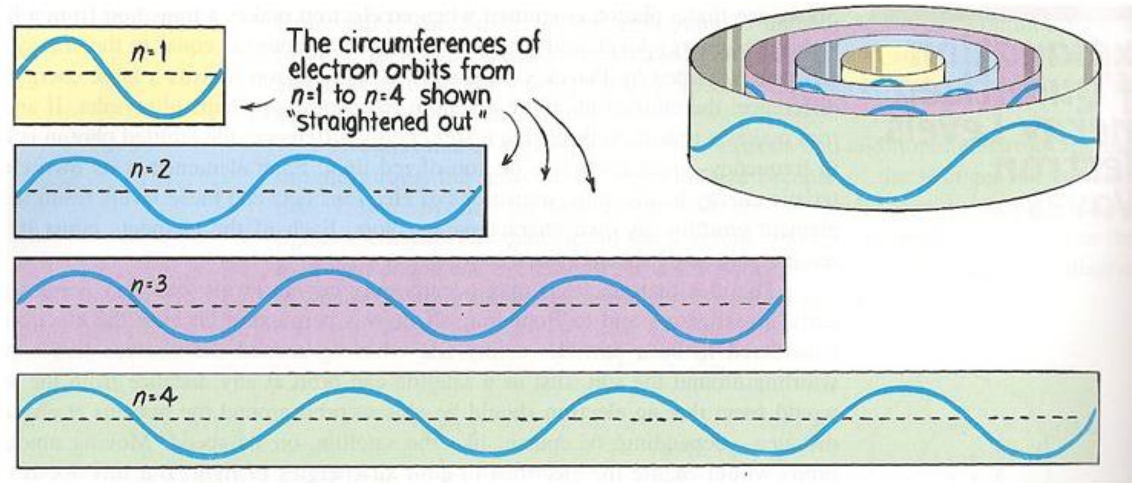
1. Wavefunction and its interpretation

2. Quantum measurement
   - Observables and operators
   - Measurement and collapse of wave function
   - Uncertainty principle

3. Time evolution of wavefunction: Schrödinger equation

Wavefunction

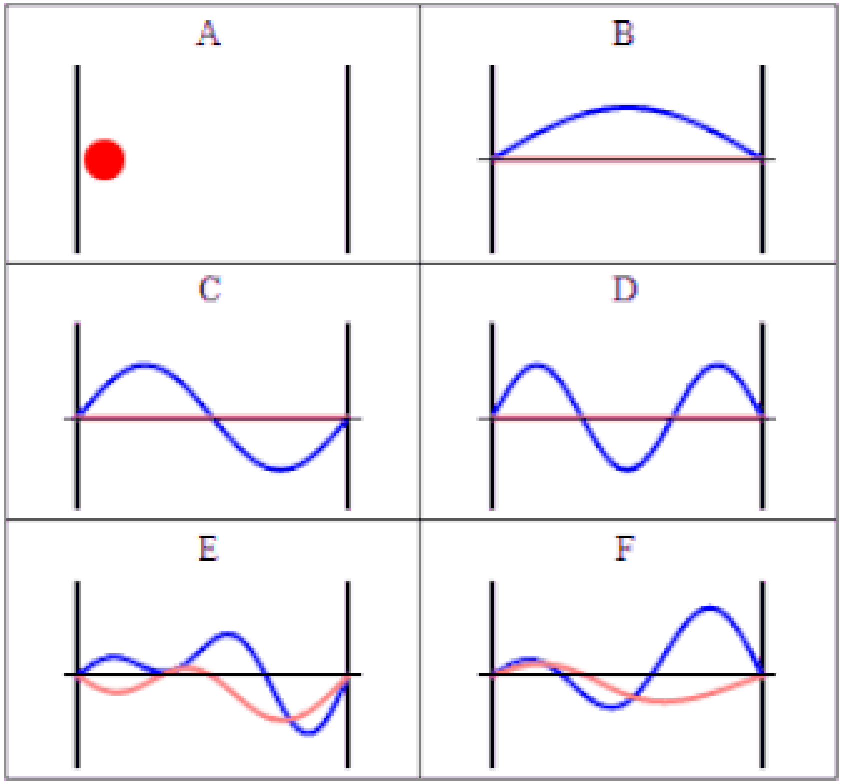
The state of a quantum-mechanical particle is represented by a certain complex function that behaves like a wave



(Image: https://chemistry.stackexchange.com/questions/116840/why-do-solutions-of-electron-in-a-box-and-in-a-ring-predict-coefficients-for-l)

Not in text

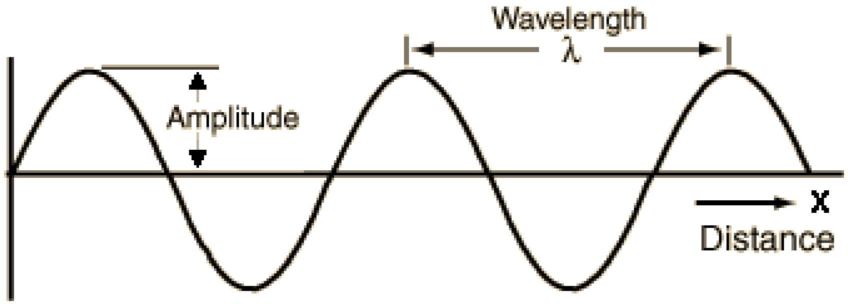
Example of a Wavefunction



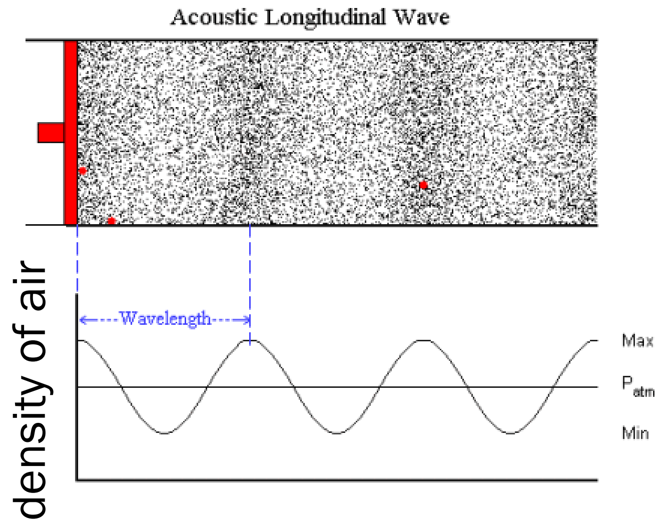
(Image: https://commons.wikimedia.org/wiki/File:InfiniteSquareWellAnimation.gif)

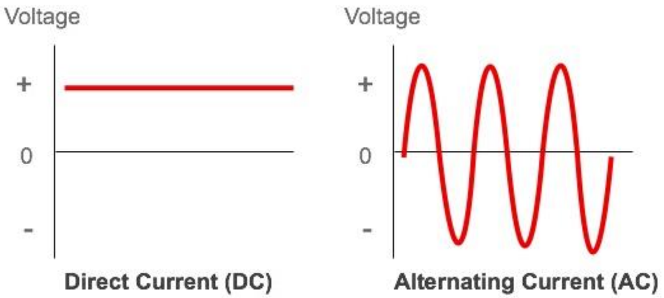
"Amplitude" of a Wave

Some value assigned for each point of the space



0 +1 +2 +1 0 -1 -2 -1 0 +1 +2 +1





RS 2.2
M 4.1

"Amplitude" of a Wavefunction

For a wavefunction $\Psi(x, t)$,

its square $|\Psi(x, t)|^2$ gives the probability

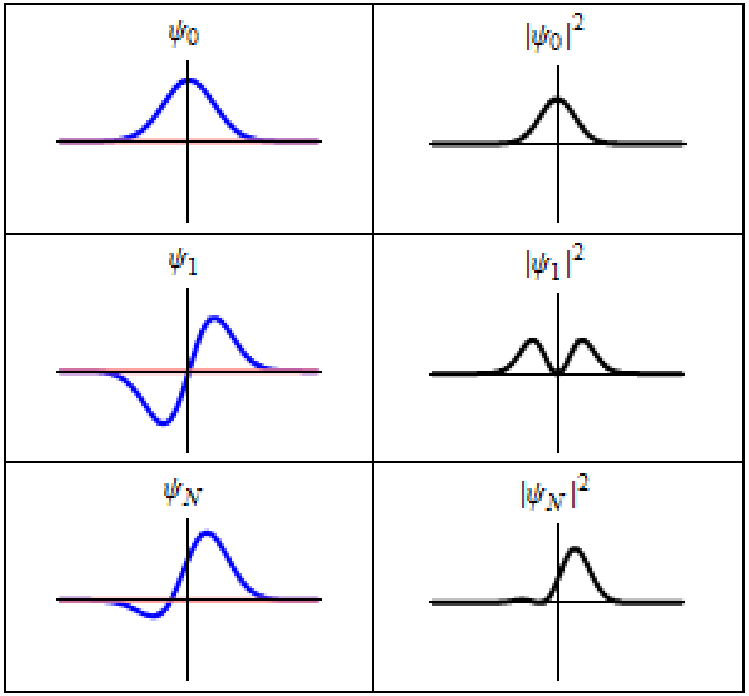
of finding the particle at point x at time t

(Born's statistical interpretation)

What really matters is the PROBABILITY!

Not in text

Wavefunction and Probability



(Image: https://www.pinterest.co.kr/pin/661818107719334402/)

RS 2.2
M 4.1

Normalization

The integral of the probability must be one:

$$\int_{-\infty}^{\infty} |\Psi(x,t)|^2 \, \mathrm{d}x = 1$$

(Modified) Example 4-1

A wave function has the form of

$$\Psi_n(x,t) = \left(\frac{2}{a}\right)^{1/2} \sin\left(\frac{\pi x}{a}\right) e^{-2\pi i E t/h}, \qquad 0 \le x \le a.$$

Show that the wave function is normalized.

The probability density is

$$|\Psi_n(x,t)|^2 = \Psi_n(x,t)\Psi_n^*(x,t) = \frac{2}{a}\sin^2\left(\frac{\pi x}{a}\right)$$

The normalization condition leads to

$$\int_{-\infty}^{\infty} |\Psi_n(x,t)|^2 \,\mathrm{d}x = \frac{2}{a}\int_0^a \sin^2\left(\frac{\pi x}{a}\right)\mathrm{d}x = \frac{2}{a} \times \frac{a}{2} = 1$$

RS 2.2
M 4.1

Once We Know a Wavefunction…

A wavefunction contains all information

about the state of the system

Quantum-mechanical measurement?

RS 2.3
M 4.2

Observables and Operators

- Every CM observable has a corresponding QM operator
- An operator performs a mathematical operation on the following function
  - *e.g.* "+3" operator, "$\times 2$" operator, "$\mathrm{d}/\mathrm{d}x$" operator
  - Typically denoted by a letter with a caret ($\hat{}$)

RS 2.3
M 4.2

Observables and Operators

- Every CM observable has a corresponding QM operator

- Position operators: $\hat{x} = x,\ \hat{y} = y,\ \hat{z} = z$
- Momentum operators:

$$\hat{p}_x = \frac{\hbar}{i}\frac{\partial}{\partial x}, \qquad \hat{p}_y = \frac{\hbar}{i}\frac{\partial}{\partial y}, \qquad \hat{p}_z = \frac{\overset{h/2\pi}{\overset{\downarrow}{\hbar}}}{i}\frac{\partial}{\partial z}$$

- System energy operator (Hamiltonian):

$$\hat{H} = -\frac{\hbar^2}{2m}\nabla^2 + \mathcal{V}(\hat{x}, \hat{y}, \hat{z})$$

RS 2.3
M 4.2

Eigenvalue Equation

For a given operator $\hat{A}$, a function f such that

$$\hat{A}f = af$$

is called an eigenfunction of $\hat{A}$, and a is called an eigenvalue. This equation is called an eigenvalue equation.

e.g. For $\hat{A} = \mathrm{d}/\mathrm{d}x$,

e^{2x} is an eigenfunction with the eigenvalue of 2.

RS 2.3
M 4.2

Properties of QM Operators

- Eigenvalues are real
- Eigenfunctions form a complete set

  "Any arbitrary function can be represented as a linear combination of the functions from a complete set"

- Eigenfunctions are orthogonal to each other

  "Two functions are orthogonal to each other if their inner product is zero"

$$\int f^* g \, \mathrm{d}\tau = 0$$

RS 2.5
M 4.3

Measurement in QM

If a wavefunction Ψ of the system satisfies

$$\hat{A}\Psi = a\Psi$$

for a certain operator $\hat{A}$, the measurement of the corresponding observable always gives a (eigenvalue).

Not in text

Example

The x-axis momentum operator is

$$\hat{p}_x = \frac{\hbar}{i}\frac{\partial}{\partial x}.$$

If a wavefunction of the system is

$$\Psi(x) = e^{ikx},$$

the following eigenvalue equation holds:

$$\hat{p}_x \Psi(x) = (\hbar k)\Psi(x).$$

Hence, measurement of the system momentum always gives $\hbar k$.

M 4.8

If Wavefunction ≠ Eigenfunction?

If a wavefunction is represented as a linear combination

of the eigenfunctions of an operator ($\hat{A} f_k = a_k f_k$),

$$\Psi = c_1 f_1 + c_2 f_2 + c_3 f_3 + \cdots + c_n f_n,$$

the measurement gives the eigenvalue a_i with

the probability of the squared coefficient $|c_i|^2$.

Not in text

Example

The x-axis momentum operator $\hat{p}_x = \frac{\hbar}{i}\frac{\partial}{\partial x}$:

eigenfunction e^{ikx} with eigenvalue $\hbar k$

If a wavefunction of the system is

$$\Psi(x) = \frac{\sqrt{2}}{\sqrt{3}}e^{ix} + \frac{1}{\sqrt{3}}e^{-2ix},$$

the measurement of the system momentum gives

- $\hbar$ with the probability of $2/3$; and
- $-2\hbar$ with the probability of $1/3$

M 4.10

Collapse of a Wavefunction

When you obtain a certain value from measurement,

the wavefunction becomes the corresponding eigenfunction.

e.g. For $\Psi(x) = \frac{\sqrt{2}}{\sqrt{3}} e^{ix} + \frac{1}{\sqrt{3}} e^{-2ix}$,

if the momentum measurement gives

- $\hbar \longrightarrow$ the wavefunction becomes $\Psi(x) = e^{ix}$
- $-2\hbar \longrightarrow$ the wavefunction becomes $\Psi(x) = e^{-2ix}$

Not in text

Quantum Measurement

- If the wavefunction is the eigenfunction of the operator

$$\boxed{\Psi = e^{ix}} \xrightarrow[\hbar\ (100\%)]{\text{Meas.}} \boxed{\Psi = e^{ix}} \xrightarrow[\hbar\ (100\%)]{\text{Meas.}} \boxed{\Psi = e^{ix}}$$

- Otherwise

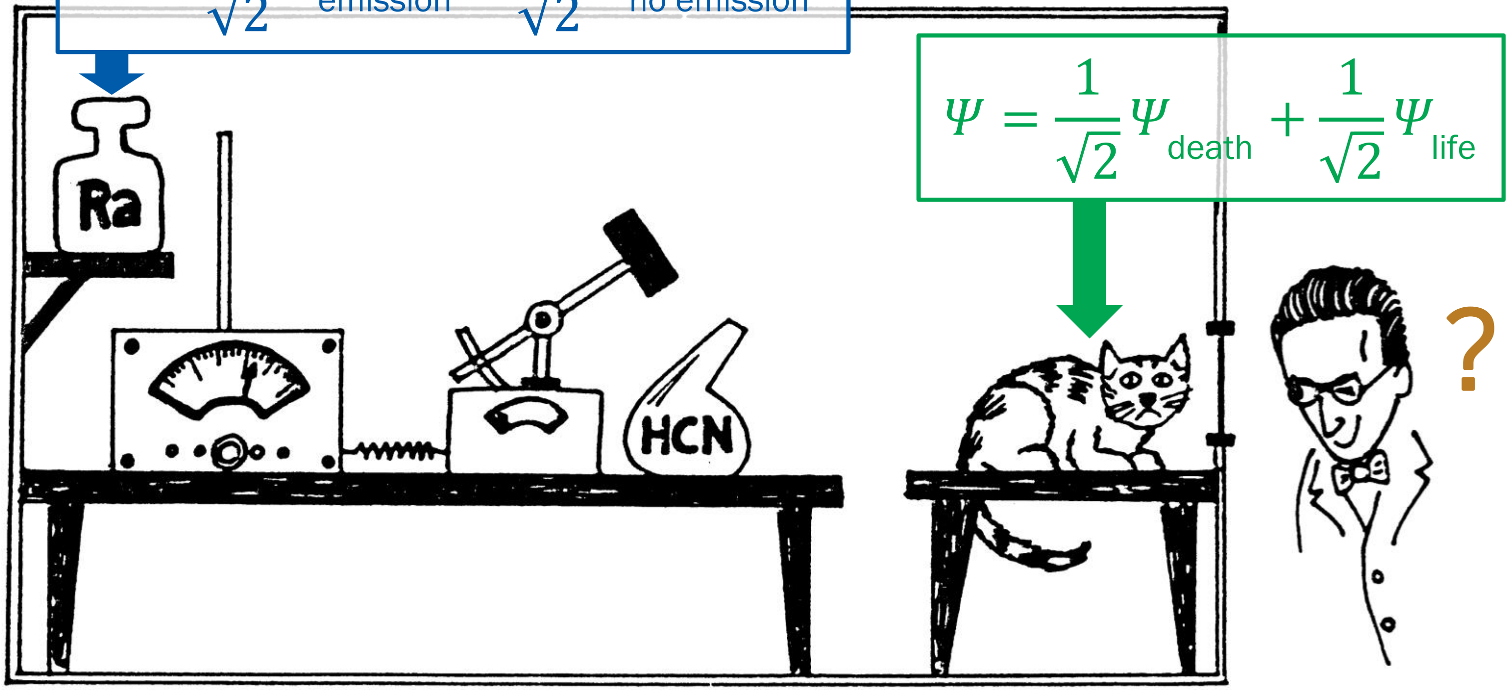
$$\begin{tikzcd}
 & \boxed{\Psi = e^{ix}} \arrow[r, "\hbar", "(100\%)"'] & \boxed{\Psi = e^{ix}} \\
\boxed{\Psi = \frac{1}{2} e^{ix} + \frac{1}{2} e^{-ix} + \frac{1}{\sqrt{2}} e^{2ix}} \arrow[ur, "\hbar\ (25\%)"] \arrow[r, "-\hbar", "(25\%)"'] \arrow[dr, "2\hbar\ (50\%)"'] & \boxed{\Psi = e^{-ix}} \arrow[r, "-\hbar", "(100\%)"'] & \boxed{\Psi = e^{-ix}} \\
 & \boxed{\Psi = e^{2ix}} \arrow[r, "2\hbar", "(100\%)"'] & \boxed{\Psi = e^{2ix}}
\end{tikzcd}$$

Not in text

Schödinger's Cat

$$\Psi = \frac{1}{\sqrt{2}}\Psi_{\text{emission}} + \frac{1}{\sqrt{2}}\Psi_{\text{no emission}}$$

$$\Psi = \frac{1}{\sqrt{2}}\Psi_{\text{death}} + \frac{1}{\sqrt{2}}\Psi_{\text{life}}$$



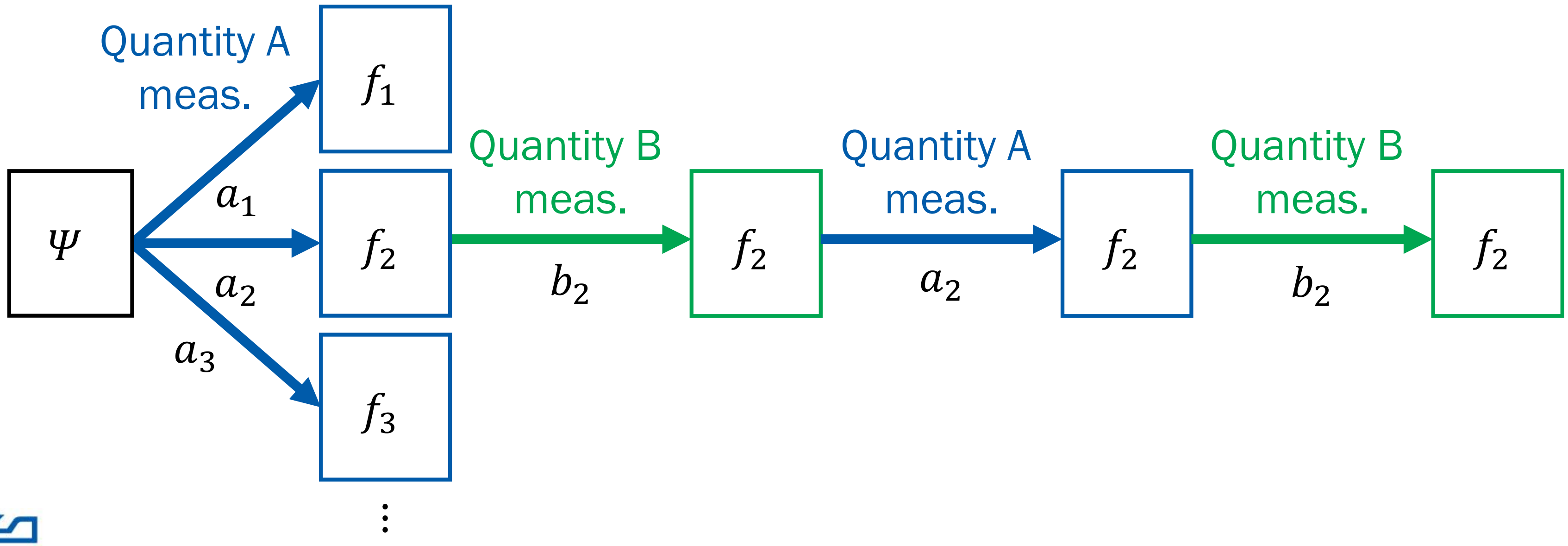
(Image: http://www.inven.co.kr/board/webzine/2097/295689)

M 4.7

Measurement of Two Observables

If two QM operators share their eigenfunctions …

$\hat{A} f_k = a_k f_k$ and $\hat{B} f_k = b_k f_k$

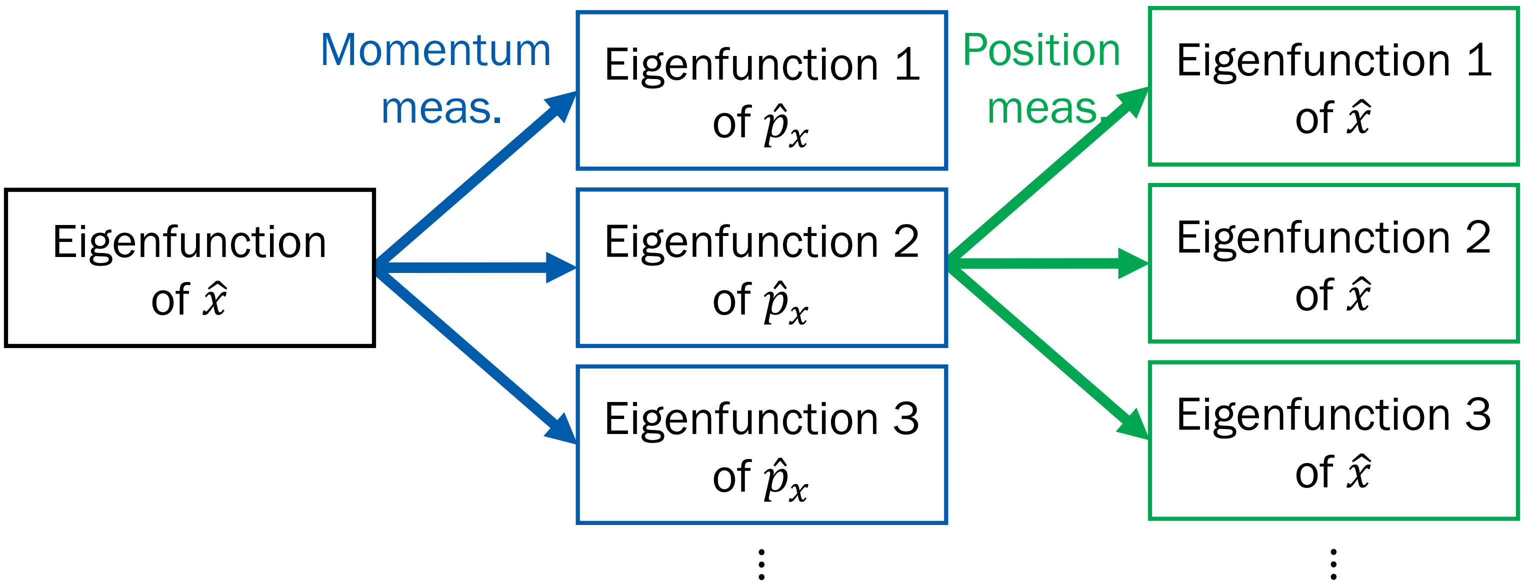


M 4.4

Measurement of Two Observables

The eigenfunction of one QM operator may not be the eigenfunction of another QM operator

(e.g. eigenfunction of $\hat{x} \neq$ eigenfunction of $\hat{p}_x$)



Uncertainty Principle

For two QM operators that do not share the eigenfunctions,

It is impossible to simultaneously determine

the corresponding observables

- Position-momentum uncertainty (Gen Chem version)

$$\Delta x \Delta p_x \geq \frac{\hbar}{2}$$

Review: Statistics

- Expected value

$$\langle g \rangle = \sum_{i=1}^{\infty} p_i g_i$$

- Variance

$$\sigma_g^2 = \langle (g - \langle g \rangle)^2 \rangle = \langle g^2 \rangle - \langle g \rangle^2$$

- Standard deviation

$$\sigma_g = \left(\sigma_g^2\right)^{1/2} = \sqrt{\langle g^2 \rangle - \langle g \rangle^2}$$

RS 2.6
M 4.3

Expectation Value

For the eigenfunctions of the operator $\hat{A}$,

$$\hat{A} f_k = a_k f_k,$$

if the wavefunction of the system is

$$\Psi = c_1 f_1 + c_2 f_2 + c_3 f_3 + \cdots + c_n f_n,$$

the expectation value of the observable A is

$$\begin{aligned} \langle A \rangle &= a_1 P_1 + a_2 P_2 + \cdots + a_n P_n \\ &= a_1 |c_1|^2 + a_2 |c_2|^2 + \cdots + a_n |c_n|^2 \end{aligned}$$

RS 2.6
M 4.3

Expectation Value

$$\langle A \rangle = a_1|c_1|^2 + a_2|c_2|^2 + \cdots + a_n|c_n|^2$$

Using the orthogonality (and normalization), we can show

$$\langle A \rangle = \int \Psi^* \hat{A} \Psi \, \mathrm{d}\tau$$

Example 4-4

Calculate the average energy of the system in a state represented by

$$\Psi(x) = \left(\frac{30}{a^5}\right)^{1/2} x(a-x), \qquad 0 \le x \le a$$

The average energy is the expectation value of energy:

$$\langle E \rangle = \int_0^a \Psi^*(x)\hat{H}\Psi(x)\,\mathrm{d}x = \int_0^a \Psi^*(x)\left(-\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2}\right)\Psi(x)\,\mathrm{d}x$$

$$\langle E \rangle = -\frac{15\hbar^2}{ma^5}\int_0^a x(a-x)\frac{\mathrm{d}^2}{\mathrm{d}x^2}[x(a-x)]\,\mathrm{d}x$$

$$\langle E \rangle = \frac{30\hbar^2}{ma^5}\int_0^a x(a-x)\,\mathrm{d}x = \frac{5\hbar^2}{ma^2}$$

RS 2.6
M 4.3

Variance

$$\sigma_A^2 = \langle A^2 \rangle - \langle A \rangle^2$$

where

$$\langle A \rangle = \int \Psi^* \hat{A} \Psi \, \mathrm{d}\tau$$

$$\langle A^2 \rangle = \int \Psi^* \hat{A} \hat{A} \Psi \, \mathrm{d}\tau$$

- Standard deviation: $\sigma_A = \sqrt{\langle A^2 \rangle - \langle A \rangle^2}$

Review

Operator Multiplication

- Associative property

$$(\hat{A}\hat{B})\hat{C} = \hat{A}(\hat{B}\hat{C})$$

- Distributive property

$$\hat{A}(\hat{B} + \hat{C}) = \hat{A}\hat{B} + \hat{A}\hat{C}$$

- Commutative property

$$\hat{A}\hat{B} = \hat{B}\hat{A}\ ? \quad \rightarrow \text{Not necessarily}$$

More on Commutative Property

- If $\hat{A}\hat{B} = \hat{B}\hat{A}$, $\hat{A}$ and $\hat{B}$ are said to commute

- Commutator

$$[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}$$

- If $\hat{A}$ and $\hat{B}$ commute,

$$[\hat{A}, \hat{B}] = \hat{0}$$

M 4.4

Generalized Uncertainty Principle

$$\sigma_A^2 \sigma_B^2 \geq -\frac{1}{4}\langle[\hat{A}, \hat{B}]\rangle^2$$

e.g. position-momentum uncertainty

$$[\hat{x}, \hat{p}_x] = \hat{x}\hat{p}_x - \hat{p}_x\hat{x} = x\frac{\hbar}{i}\frac{\partial}{\partial x} - \frac{\hbar}{i}\frac{\partial}{\partial x}x = i\hbar$$

$$\sigma_x^2 \sigma_{p_x}^2 \geq -\frac{1}{4}\langle i\hbar\rangle^2 = \left(\frac{\hbar}{2}\right)^2$$

Not in text

Energy-Time Uncertainty

$$\Delta E \Delta t \geq \frac{\hbar}{2}$$

"the energy of a system that has existed only for a finite time Δt has a spread of at least ΔE."

RS 2.4
M 4.9

Time Evolution of Wavefunction

$$i\hbar \frac{\partial}{\partial t}\Psi(x, y, z, t) = \hat{H}\Psi(x, y, z, t)$$

where

$$\hat{H} = -\frac{\hbar^2}{2m}\nabla^2 + \mathcal{V}(\hat{x}, \hat{y}, \hat{z}, t)$$

(Time-dependent Schrödinger equation)

RS 2.4
M 4.9

Time-Independent Potential

$$i\hbar \frac{\partial}{\partial t}\Psi(x,y,z,t) = \hat{H}\Psi(x,y,z,t)$$

If the potential is independent of time, we assume

$$\Psi(x,y,z,t) = \psi(x,y,z)T(t)$$

to obtain

$$\hat{H}\psi(x,y,z) = E\psi(x,y,z)$$

(time-independent Schrödinger equation)

and $T(t) = e^{-iEt/\hbar}$

RS 2.4
M 4.9

Time-Independent Solution

- Time-independent Schrödinger equation

$$\hat{H}\psi(x,y,z) = E\psi(x,y,z)$$

$\longrightarrow$ Eigenvalue equation for the Hamiltonian

- Temporal solution: $T(t) = e^{-iEt/\hbar}$

$$|T(t)|^2 = 1$$

$$|\Psi(x,y,z,t)|^2 = |\psi(x,y,z)|^2$$